

WEEK 13 to 14: ORGAN SYSTEM

MUSCULAR SYSTEM

- Composed of contractile tissue to shorten (contract) and relax due to *neural or chemical stimulation*

FUNCTIONS

- 1.) Producing voluntary movements via skeletal muscles
- 2.) Maintains essential involuntary functions via smooth and cardiac muscles
- 3.) Stabilizes body during motion and rest

large - 90

small - 55

normal - 70

TYPES OF MUSCLE

1.) Cardiac muscle

- Involuntary; striated; only in heart
 - For continuous, rhythmic, and fatigue-resistant contractions
- RMP: -90 mV
- Only found in myocardium (muscular middle layer of the heart wall)

STRUCTURAL CHARACTERISTICS

1.) Striated fibers

- Sarcomere-based striations
- Contains actin, myosin, troponin, and tropomyosin in repeating units

2.) Branched fibers

- Shorter, branched, and highly interconnected
- Promoted three-dimensional contraction

3.) Single central nucleus

- Contains one centrally located nucleus

4.) Intercalated discs

- Specialized junctions between cardiac cells
 - Gap junctions - electrical transmission (syncytium)
 - Desmosomes - anchor cells during high-pressure contractions
- For rapid, coordinated contraction across myocardium

FUNCTIONAL CHARACTERISTICS

1.) Rhythmic, coordinated contraction

- Contract in syncytium, effective for pumping blood

2.) High-fatigue resistance

- Abundant mitochondria ($\approx 25\%$ of cell volume)
- Rich in myoglobin and blood supply
- Reliant on aerobic metabolism

3.) Automaticity & conductivity

- Can generate and propagate APs quickly; SA node

4.) Elasticity and tension regulation

- Frank-Starling law: greater stretch, greater contraction

2.) Smooth muscle

- Involuntary; non-striated; found in the walls of hollow organs
 - Slow, sustained, and rhythmic contractions
- RMP: -55 mV

STRUCTURAL CHARACTERISTICS

1.) Non-striated appearance

- Lacks sarcomeres; uniform ("smooth") appearance under microscope
- Contain actin and myosin but arranged diagonally rather than in repeating units

2.) Spindle-shaped cells

- Long, tapered cells with single, centrally placed nuclei
- Arranged in sheets or layers (circular & longitudinal)

3.) Dense bodies

- Functional equivalent of Z-discs; anchoring point for actin filament
- When it contracts, it shortens in a corkscrew or spiral fashion

4.) High-capacity for stretch

- Plasticity; can stretch while maintaining contractile function (bladder, uterus)

CONTROL & REGULATION

1.) Involuntary control

- Regulated by:
 - ANS
 - Sympathetic - high contraction of vessels, low GI motility
 - Parasympathetic - high GI motility
 - Hormones (oxytocin, epinephrine, estrogen)
 - Local factors (CO₂, O₂, pH, NO)

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2.) *Slow, sustained contraction*

- Slower but can be prolonged with minimal energy use
- Uses calmodulin instead of troponin for Ca regulation

3.) *Two types of smooth muscle*

- Single-unit (visceral):
 - Found in GI tract, uterus, bladder
 - Contraction as a unit due to gap junction
 - Responsible for peristalsis and rhythmic patterns
- Multi-unit:
 - Found in iris, arrector pili, large airways
 - Independently innervated -> fine control

3.) *Skeletal muscle*

- Voluntary; striated; attached to bones via tendons
 - For producing body movement, maintaining posture, stabilizing joints, and generating heat
- RMP: -90 mV
- 640 in total; 602 in old book review

STRUCTURAL CHARACTERISTICS

1.) *Striated appearance*

- Sarcomere-based striations; repeating units and actin and myosin
- Creates visible light (I-band) and dark (A-band) patterns under the microscope

2.) *Multinucleated fibers*

- Formed by the fusion of multiple myoblasts
- Nuclei are peripherally located—hallmark feature

3.) *Long-cylindrical fibers*

- Can extend the full length of a muscle
- Organized into bundles:
 - Muscle -> fascicles -> muscle fibers -> myofibrils -> myofilaments

4.) *Connective tissue layers*

- For transmitting force and structural support
 - Epimysium - surrounds the entire muscle
 - Perimysium - surrounds fascicles
 - Endomysium - surrounds individual fibers

5.) *Neuromuscular junction (NMJ)*

- Specialized synapse between motor neurons and muscle fiber
- Uses ACh to initiate contraction

MICROSCOPIC STRUCTURE

1.) *Muscle fiber* — long, cylindrical, multinucleated formed from the fusion of myoblasts

- Key characteristic:

- Multinucleated: nuclei are located at the periphery beneath the sarcolemma
- Sarcolemma: cell membrane; propagates APs
- Sarcoplasm: cytoplasm of muscle fiber; contains mitochondria, glycogen stores, myoglobin
- Length: entire length of a muscle
- Diameter: 10-100 um, but still depends

- Specialized structures:

- T-tubules — transverse tubules: invaginations of the sarcolemma
- Sarcoplasmic reticulum (SR): stores and releases Ca

2.) *Myofibrils* — long, cylindrical bundles of contractile proteins (actin & myosin)

- Composition:

- Repeating units of sarcomeres arranged end-to-end

- Function:

- Muscle contraction
- Increase in number during hypertrophy, contributing to larger muscle size -> from muscle tears in exercise

3.) *Sarcomere* — functional and structural unit of skeletal muscle contraction

- Boundaries:

- Z-line to Z-line

- Major bands and lines:

- Z-line: anchors actin
- A-band: length of myosin (dark region)
- I-band: only actin (light region)
- H-zone: center of A-band; contains only myosin
- M-line: center of H-zone; stabilize myosin

CONTROL AND REGULATION

1.) *Voluntary control*

- Innervated by somatic motor neurons
- Motor unit recruitment determines the strength

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and precision

2.) **Excitation-contraction coupling**

- AP trigger Ca release from SR; Ca binds troponin, initiating cross-bridge cycling

3.) **Motor units**

- Small motor units: fine control (hands, face)
- Large motor units: power (glutes, quadriceps)

MUSCLE ARCHITECTURE

- Arrangement of muscles fibers relative to the line of pull

- Strongest predictors of muscle function, influencing:

- Force production, ROM, contraction speed, physiological cross-sectional area (PCSA)

- **General rule:**

- More fibers in parallel -> greater force
- Longer fibers -> greater excursion & speed

1.) **Parallel — strap muscles**

- Example: sartorius, SCM, gracilis, platysma

- Characteristics:

- Long, thin fibers
- Capable of long excursions (large ROM)
- Produce fast but less powerful contraction

- Function:

- High-speed movements
- Large joint ranges (e.g., hip flexion; abduction; external rotation (ER) with sartorius)

2.) **Fusiform — spindle-shaped**

- Example: biceps brachii, brachialis, brachioradialis — elbow flexors; tibialis anterior

- Characteristics:

- Thick belly with tapered ends
- Fibers run parallel but are more concentrated in the belly, increasing force

- Function:

- Stronger than strap muscles
- Good ROM AND contraction speed
- Efficient in lifting and pulling motions

3.) **Pennate fiber**

- Fibers arranged at an angle relative to the tendon, similar to a feather; **higher PCSA** ->

greater force

- Trade off: increased force but decreased ROM & velocity

a.) Unipennate

- **Example**: extensor digitorum & hallucis longus, tibialis posterior
- **Characteristic**: fibers inserted diagonally on one side of the tendon
- **Function**:
 - Good force production; moderate ROM; efficient for repetitive, endurance-based movements

b.) Bipennate

- **Example**: rectus femoris, gastrocnemius (partially)
- **Characteristic**: fibers inserted diagonally on both sides of the tendon
- **Function**:
 - More forceful; common major locomotor muscles of the lower limb; designed for power production during gait, jumping, climbing

c.) Multipennate

- **Example**: deltoid, soleus (complex pennation)
- **Characteristics**: multiple tendinous intersections with fibers arranged in several angles
- **Function**:
 - Strongest muscle architecture; highest PCSA - greatest force generation; least ROM; ideal for stabilization and heavy lifting

3.) **Convergent — triangular**

- Example: pectoralis major, latissimus dorsi, temporalis. gluteus maximus

- Characteristics:

- Broad origin -> converge to a single tendon
- Fiber orientation varies: superior, inferior, middle

- Function:

- Allows multiple lines of pull
- Highly versatile: adduction, flexion, extension, internal rotation
- Produces powerful movements in multiple planes

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4.) Circular — sphincter muscles

- Example: orbicularis oris, orbicularis oculi, pyloric sphincter, anal sphincter

- Characteristics:

- Fibers arranged in concentric rings
- Surround body openings

- Function:

- Regulate passage (opening/closing) through orifices
- Important in eye blinking, mouth closure, speech, facial expression, continence (anal/urethral sphincter)

4.) *Serratus anterior* - serrated, saw-like

4.) Orientation of fibers

EXAMPLES

- 1.) *Rectus* - straight (rectus abdominis)
- 2.) *Oblique* - diagonal (external & internal obliques) — reversed functions
- 3.) *Transverse* - horizontal (transversus abdominis)

5.) Action

EXAMPLES

- 1.) *Flexor* - flexor carpi radialis
- 2.) *Extensor* - extensor digitorum
- 3.) *Adductor* - adductor longus
- 4.) *Abductor* - abductor pollicis longus
- 5.) *Pronator* - pronator teres
- 6.) *Supinator* - supinator

6.) Number of heads

EXAMPLES

- 1.) *Biceps* - two heads (biceps brachii, biceps femoris)
- 2.) *Triceps* - three heads (triceps brachii)
- 3.) *Quadriceps* - four heads (quadriceps femoris group)

CLINICAL RELEVANCE

1.) Multiple heads

- Often cross multiple joints
- Have multiple actions
- Provide stability through redundancy

7.) Attachments

EXAMPLES

1.) SCM

- Origin: sterno- (sternum) + cleido- (clavicle)
- Insertion: mastoid process of temporal bone

2.) Coracobrachialis

- Coracoid process -> humerus

3.) Iliocostalis

MUSCLE NAMING

1.) Location

EXAMPLES

- 1.) *Brachialis* - brachium (arm)
- 2.) *Femoris* (e.g., *rectus femoris*) - femoral region (thigh)
- 3.) *Pectoralis* - chest
- 4.) *Intercostal* - between ribs

2.) Size

- Often use Latin terms
- Larger muscles = power; longer muscles = excursion

KEY TERMS

- 1.) *Maximus* - largest (gluteus maximus)
- 2.) *Minimus* - smallest (gluteus minimus)
- 3.) *Medius* - middle-sized (gluteus medius)
- 4.) *Longus* - long (adductor longus)
- 5.) *Brevis* - short (peroneus brevis)
- 6.) *Major/magnus* - large (pectoralis major, adductor magnus)
- 7.) *Minor* - small (pectoralis minor)

3.) Shape

EXAMPLES

- 1.) *Deltoid* - triangular
- 2.) *Trapezius* - trapezoid-shaped
- 3.) *Rhomboids* - rhombus-shaped

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- Ilium -> ribs

4.) **Stylohyoid**

- Styloid process -> hyoid bone

MAJOR MUSCLE REGIONS

HEAD AND FACE

1.) Scalp — forehead region

1.) **Occipitofrontalis — epicranius**

- **Frontal belly (frontalis)** - raises eyebrows, wrinkles forehead
- **Occipital belly (occipitalis)** - retracts scalp
- **Connected by:** galea aponeurotica

2.) **Temporoparietalis**

- Assists in moving scalp and ears

2.) Periorbital — around the eye

1.) **Orbicularis oculi**

- Closes eyelids (blinking, squinting)
- In bell's palsy, eyes are open

2.) **Corrugator supercilii**

- Draws eyebrows medially & downward (frowning)

3.) **Levator palpebrae superioris**

- Opens eyelids
- Innervation: CN2

4.) **Depressor supercilii**

- Assists in lowering the eyebrow

3.) **Nasal region**

1.) **Nasalis (2 parts)**

- **Transverse part** - compresses nasal aperture
- **Alar part** - flares nostrils

2.) **Procerus**

- Wrinkles skin over nose bridge (anger expression)

3.) **Depressor septi nasi**

- Pulls nasal septum downward, assists in narrowing nostrils

4.) **Oral region**

1.) **Orbicularis oris**

- Closes and protrudes lips (kissing)

2.) **Buccinator**

- Compresses cheeks (blowing, chewing), prevents food from pocketing

3.) **Risorius**

- Draws corners of mouth laterally (wide smile)

4.) **Zygomaticus major & levator anguli oris**

- Elevates corner of mouth (smiling)

5.) **Zygomaticus minor & levator labii superioris**

- Elevates upper lip

6.) **Levator labii superioris alaeque nasi**

- Elevates upper lip and flares nostrils (Elvis muscle)

7.) **Depressor anguli oris**

- Depresses corner of mouth (frowning)

8.) **Depressor labii inferioris**

- Lowers bottom lip

9.) **Mentalis**

- Protrudes lower lip (pouting), wrinkles chin

5.) **Auricular muscles — around the ear**

1.) **Auricularis anterior**

- Moves ear forward

2.) **Auricularis superior**

- Elevates the ear

3.) **Auricularis posterior**

- Retracts ear

6.) **Neck region — affects facial expression**

1.) **Platysma**

- Tenses skin of neck, depresses jaw & lower lip
- Important in expressions of stress and fright

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7.) Neck muscles

SUPERFICIAL LAYER

1.) Sternocleidomastoid (SCM)

- Action: bilateral: neck flexion; unilateral: rotates head to opposite side, lateral flexion
- Nerve: CN11 (motor), C2-C3 (sensory)

SUPRAHYOID MUSCLES

Function: elevate hyoid, assist in swallowing & speaking, open mouth (depress mandible)

1.) Digastric — anterior belly

- Action: elevates hyoid; depresses mandible

2.) Digastric — posterior belly

- Action: elevates hyoid

3.) Mylohyoid

- Action: elevates hyoid & floor of mouth

4.) Stylohyoid

- Action: elevates & retracts hyoid

5.) Geniohyoid

- Action: pulls hyoid anterosuperiorly; widens pharynx

INFRAHYOID MUSCLES — TOSS MUSCLE

Function: depress or stabilize hyoid during swallowing and speech

1.) Thyrohyoid

- Action: elevates thyroid cartilage or depresses hyoid

- Nerves: C1 via CN12

2.) Omohyoid — superior & inferior bellies

- Action: depresses, retracts, stabilizes hyoid

- Nerve: ansa cervicalis

3.) Sternohyoid

- Action: depresses hyoid

- Nerve: ansa cervicalis (C1-C3)

4.) Sternothyroid

- Action: depresses thyroid cartilage

- Nerve: ansa cervicalis

8.) Trunk muscles

THORACIC MUSCLES

1.) External intercostalis

- Action: elevate ribs -> inspiration
- Nerve: intercostal nerves (T1-T11)

2.) Internal intercostalis

- Action: depress ribs -> forced expiration

- Nerve: intercostal nerves

3.) Innermost intercostals

- Action: assist internal intercostals

- Nerve: intercostal nerves

4.) Diaphragm

- Action: primary muscle of respiration; contracts & flattens during inspiration

- Nerve: phrenic nerve (C3-C5)

ABDOMINAL MUSCLES

1.) Rectus abdominis

- Action: trunk flexion; compress abdomen

- Nerve: intercostal nerves (T7-T12)

2.) External oblique

- Action: contralateral rotation; trunk flexion; compress abdomen

- Nerve: T7-T12

3.) Internal oblique

- Action: Ipsilateral rotation; trunk flexion; compress abdomen

- Nerve: T7-L1

4.) Transversus abdominis

- Action: Core stabilization; compress abdomen

- Nerve: T7-L1 (iliohypogastric, ilioinguinal nerves)

- Linea alba - separation of abdomen

9.) Back muscles

SUPERFICIAL LAYER — SCAPULAR MOVEMENT

1.) Trapezius

- Action: elevates, depresses, retracts scapula; upward rotation of scapula

- Nerve: CN11 (motor), C3-C4 (sensory)

2.) Latissimus dorsi

- Action: shoulder extension, adduction, internal rotation

- Nerve: thoracodorsal nerve (C6-C8)

3.) Rhomboid major/minor

- Action: retract & downwardly rotate scapula

- Nerve: dorsal scapular nerve (C4-C5)

4.) Levator scapulae

- Action: elevates scapula; downward rotation

- Nerve: dorsal scapular nerve (C5), C3-C4

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DEEP LAYER — POSTURE & SPINAL MOVEMENT

1.) Erector spinae (ILS)

- Subcomponents: iliocostalis, longissimus, spinalis
- Action: extension & lateral flexion of spine
- Nerve: dorsal rami of spinal nerves

2.) Transversospinalis group

- Subcomponents: multifidus, rotatores, semispinalis
- Action: stabilization; rotation; extension
- Nerve: dorsal rami of spinal nerves

MUSCLE NICKNAME

- 1.) Boxer's muscle - serratus anterior
- 2.) Climbing muscle - latissimus dorsi
- 3.) Tailor's muscle - sartorius
- 4.) Spurt muscle - brachialis
- 5.) Shunt muscle - brachioradialis
- 6.) Forgotten muscle - subscapularis
- 7.) Key muscle - piriformis
- 8.) Grinning muscle - risorius
- 9.) Smiling muscle - zygomaticus major
- 10.) Grief muscle - depressor anguli oris
- 11.) Locking muscle - quadriceps femoris
- 12.) Running muscle - gastrocnemius
- 13.) Kissing muscle - orbicularis oris
- 14.) Boxer's friend - deltoid